

CompTIA.

A+ Core 1

Exam 220-1201

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Module 1



What Does an IT Specialist Do?

Prepare for A+ Certification by:

- Describing what an IT specialist is and their responsibilities
- Describing the skills an IT specialist needs
- Describing the role of certifications for an IT specialist

Lesson 1.1

The Hero of Problem Solving

Role of an IT Specialist



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- Problem Solver
 - Responsibilities
- Range of Issues
 - Ever-changing daily tasking- No routine

Skills and Abilities (Slide 1 of 2)

- Problem Solving
 - Use a consistent process
- Communication
 - Written
 - Oral
 - Effective
- Organization
 - Clean workspace = easier to locate tools and parts
 - Organize thoughts that then are communicated to stakeholders



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Skills and Abilities (Slide 2 of 2)

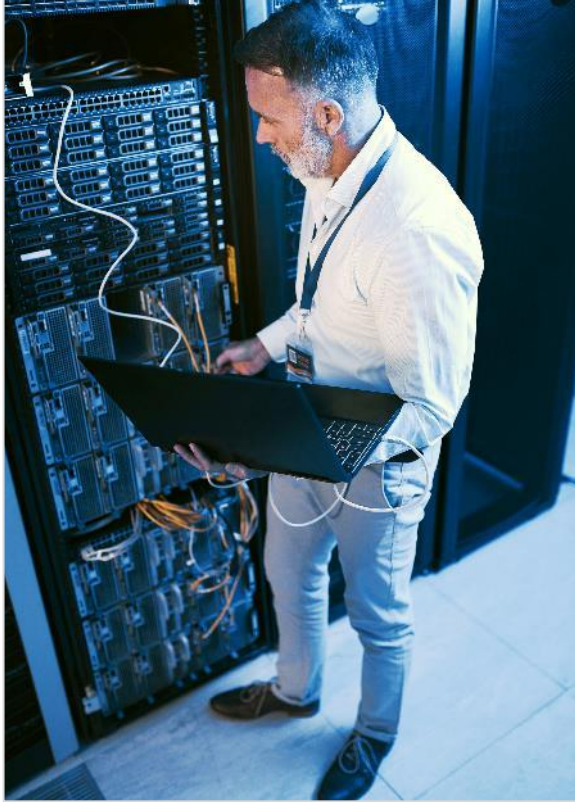


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- Technical Knowledge
 - Hardware
 - Software- including operating systems and applications
 - Networking
 - Cloud
 - Security

Lesson Review: The Hero of Problem Solving



Lesson 1.2

The Troubleshooting Methodology

Best Practice Methodology

Determine cause, symptoms and consequences

Priority and severity

- Critical systems and most severe issues need to be addressed first

CompTIA Troubleshooting Methodology

- Identify the problem
- Establish a theory of probable cause
- Test the theory
- Establish a plan of action and implement the solution
- Verify full system functionality
- Document findings, actions, outcomes

Identify the Problem

Gather information from the user

- What are the exact error messages appearing on the screen or coming from the speaker?
- Is anyone else experiencing the same problem?
- How long has the problem been occurring?
- What changes have been made recently to the system? Were these changes initiated by you or via another support request?

Perform backups

- Backup all critical and important information
- Verify customer or client has done so if they are responsible for the system

Establish and Test a Theory

- Conduct research
 - Physically inspect the system
 - Reproduce the problem
 - Check system documentation or owner's manual
 - Ask other technicians for their thoughts; talk to previous technician who have worked on the system
 - Read vendor documentation and use the internet to conduct research for possible causes and solutions

Question the Obvious

1

Never overlook a simple solution such as ensuring the item is turned on or plugged in

2

Decide if the issue is likely hardware or software related

3

Determine if the issue is physical or a logical

4

Test each theory individually to isolate the problem

Lesson Review: Establish a New Theory or Escalate



If initial theory does not check out, explore other causes and theories



Escalate the issue if it is:

Beyond your skill level

Beyond your knowledge level

Out of scope per company policy or warranty documentation

Implement a Plan of Action

Repair, replace, workarounds

- Determine if solution to repair the system is cost effective
- In some cases, may be cheaper to replace
- Not all issues required to be resolved, in some cases a work around may be feasible

Implement the solution

- Apply the fix or replace the item
- If a workaround is to be put in place, ensure it will work correctly

Verify and Document

Verify solution
resolves the
issue

- Ensure that the system is fully functional
- Implement preventative measures to prevent reoccurrence

Document
findings, actions,
and outcomes.

- Parts used- cost and point of purchase information
- Steps taken- be as thorough as possible; more detail the better
- Tools used- software related or physical tools that were helpful to resolve the issue
- Lessons learned



Lesson Review: The Troubleshooting Methodology

